

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Amazon E-Commerce Order Management System. The system manages customers, sellers, products, orders, payments, and deliveries while ensuring secure and efficient order processing.

1.2 Project Scope

The system allows customers to register, browse products, add items to a cart, place orders, make secure payments, track deliveries, manage returns and refunds, and maintain complete order history.

1.3 Intended Users

- Customers
- Sellers
- Warehouse Staff
- Delivery Partners
- Customer Support Executives
- Administrators

2. Business Requirements

- Secure customer registration and login
- Maintain customer profiles and multiple delivery addresses
- Manage product catalog and inventory

- Process online payments
- Track orders in real time
- Support returns and refunds
- Generate sales and inventory reports
- Send order notifications via email/SMS

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	Register customers
FR-2	Customer login.
FR-3	Search products
FR-4	Shopping cart management
FR-5	Place orders.
FR-6	Secure payment processing.
FR-7	Order tracking (Ordered → Packed → Shipped → Out for Delivery → Delivered)
FR-8	Return and refund requests
FR-9	View order history
FR-10	Admin sales reports

4. Non-Functional Requirements

- Response time under 2 seconds
- Encrypted passwords and secure payment information
- 99.9% system availability
- Scalable for millions of customers
- Daily backup and disaster recovery

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
customer_id	INT (Primary Key)	Unique identifier for each customer
full_name	VARCHAR(100)	Customer's full name
email	VARCHAR(100)	Unique email used for login/contact
phone_number	VARCHAR(15)	Customer contact number
address	VARCHAR(255)	Shipping address of the customer
registration_date	DATE	Date the customer account was created

6. Assumptions and Constraints

- Only registered customers can place orders.
- Every order belongs to one customer.
- Products must be in stock before purchase.
- Passwords are stored in encrypted form.

7. Conclusion

This document defines the Week 1 requirements for the Amazon E-Commerce Order Management System with a focus on the Customer Database Module. It provides the foundation for designing the ER diagram, database schema, and future modules such as Products, Orders, Payments, Inventory, Delivery, and Returns.